

Closing the literature–simulation loop: an untiring, citation-grounded agent for catalyst discovery

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Précis [1] is an open-source agentic research system built as an *untiring collaborator*: it never sleeps, works from real feedback, and lets no claim stand without a precise citation — a specific paragraph in a real publication that supports it, so hallucination is designed out rather than apologized for. Papers arrive automatically from open-access sources and continuous citation-graph exploration; the corpus currently holds over 22,000 fully ingested papers (33,000 tracked), broken into roughly 3 million searchable chunks. Precomputed keywords, summaries, and a clustered dynamic table of contents let language-model agents locate very specific claims token-efficiently instead of reading whole papers, and tasks can park themselves until a paper is ingested or a human answers, then resume on their own.

Discovery runs as a perpetual *literature–simulation loop*. A discovery agent owns all chemistry and holds one proposal in flight; each candidate is priced by a progressive-fidelity simulation ladder — a relax-only thermodynamic screen, then climbing-image nudged-elastic-band barriers [2] under machine-learned interatomic potentials (MACE [3], CHGNet [4]), then a verify tier — under capped budgets, and results rank on a Pareto frontier over the campaign’s objective axes (barrier, material cost, stability). The literature sits *inside* the loop, not beside it: every conjecture is minted as a typed hypothesis finding, and between compute steps the corpus accumulates cited evidence for and against it, so the next proposal is informed by what the field has already measured. A refuted hypothesis converts into a negative ruling — cited, searchable, and excluded from re-proposal by default — so the loop cannot re-spend compute on ideas the literature or its own simulations have already killed.

The loop is also honest about its own measurements. Untrusted results — unconverged barrier searches, detached adsorbates, symmetry-equivalent candidates whose barriers disagree — can never rank or graduate, but stay visible as provisional bands, so the agent is never blind to what it has computed. The governing design principle is to substitute thinking for compute: one hour at the library is worth a month in the simulator, so work runs as tight idea–probe–evaluate cycles in which every simulation is argued with the literature before it is bought and interpreted against it afterwards. We demonstrate the loop on $\text{NO} \rightarrow \text{NH}_3$ over Pd-based catalysts (Figure 1).

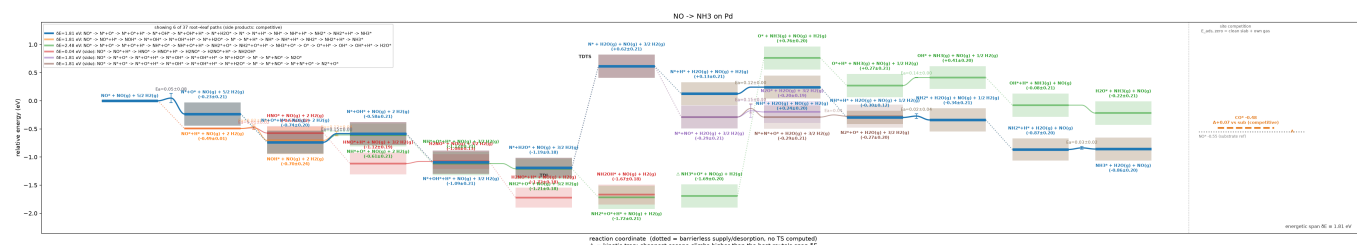


Figure 1: Reaction energy profile for $\text{NO} \rightarrow \text{NH}_3$ on Pd from the loop’s simulation tier (demonstration potential): span-ranked competing routes at full gas composition, transition states with E_a and uncertainty bands, kinetic-trap flags, and the CO site-competition panel.

References

- [1] R. Stamm, *Précis*, <https://github.com/retospect/precis-mcp> (2026).
- [2] G. Henkelman, B. P. Uberuaga, and H. Jónsson, *J. Chem. Phys.* **113**, 9901 (2000).
- [3] I. Batatia, D. P. Kovács, G. N. C. Simm, C. Ortner, and G. Csányi, *Adv. Neural Inf. Process. Syst.* **35**, 11423 (2022).
- [4] B. Deng *et al.*, *Nat. Mach. Intell.* **5**, 1031 (2023).